

## Exercises 6

**Exercise 1** (Galois Connection). Recall that a Galois connection is defined by two monotonic functions  $\alpha : C \rightarrow A$  and  $\gamma : A \rightarrow C$  between partial orders  $\leq$  on  $C$  and  $\sqsubseteq$  on  $A$ , such that

$$\forall a, c. \quad \alpha(c) \sqsubseteq a \iff c \leq \gamma(a) \quad (*)$$

1. Show that the condition  $(*)$  is equivalent to the conjunction of these two conditions:

$$\forall c. \quad c \leq \gamma(\alpha(c)) \quad (1)$$

$$\forall a. \quad \alpha(\gamma(a)) \sqsubseteq a \quad (2)$$

**Solution:**

We first show that  $(*) \implies (1)$  and  $(*) \implies (2)$

$$\begin{aligned} \alpha(c) &\sqsubseteq \alpha(c) && \text{by reflexivity of } \sqsubseteq \\ \implies c &\leq \gamma(\alpha(c)) && \text{by } (*), \text{ with } a := \alpha(c) \end{aligned}$$

and,

$$\begin{aligned} \gamma(a) &\leq \gamma(a) && \text{by reflexivity of } \leq \\ \implies \alpha(\gamma(a)) &\sqsubseteq a && \text{by } (*), \text{ with } c := \gamma(a) \end{aligned}$$

For the converse, we have:

$$\begin{aligned} \alpha(c) &\sqsubseteq a \\ \implies \gamma(\alpha(c)) &\leq \gamma(a) && \text{by monotony of } \gamma \\ \implies c &\leq \gamma(\alpha(c)) \leq \gamma(a) && \text{by (1)} \\ \implies c &\leq \gamma(a) && \text{by transitivity of } \leq \end{aligned}$$

and,

$$\begin{aligned} c &\leq \gamma(a) \\ \implies \alpha(c) &\sqsubseteq \alpha(\gamma(a)) && \text{by monotonicity of } \alpha \\ \implies \alpha(c) &\sqsubseteq \alpha(\gamma(a)) \sqsubseteq a && \text{by (2)} \\ \implies \alpha(c) &\sqsubseteq a && \text{by transitivity of } \sqsubseteq \end{aligned}$$

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2. Let  $\alpha : C \rightarrow A$  and  $\gamma : A \rightarrow C$  satisfy the conditions of a Galois connection. Show that the following three conditions are equivalent:

- (a)  $\alpha(\gamma(a)) = a$  for all  $a$
- (b)  $\alpha$  is a surjective function
- (c)  $\gamma$  is an injective function

**Solution:**

We will show that (a)  $\implies$  (b), (b)  $\implies$  (c), and (c)  $\implies$  (a).

(a)  $\implies$  (b) :

For each  $a \in A$ , there exists  $c = \gamma(a) \in C$  such that  $\alpha(c) = \alpha(\gamma(a)) = a$ . Thus,  $\alpha$  is surjective.

(b)  $\implies$  (c) :

Suppose, towards a contradiction, that  $\gamma$  is not injective. There must exist some distinct  $a_1, a_2 \in A$  such that  $\gamma(a_1) = \gamma(a_2)$ .

Further, since  $\alpha$  is surjective, there exists some  $c_1 \in C$  such that  $\alpha(c_1) = a_1$ .

$$\begin{array}{ll}
 \alpha(c_1) = a_1 & \\
 \implies \alpha(c_1) \sqsubseteq a_1 & \text{by reflexivity of } \sqsubseteq \\
 \implies c_1 \leq \gamma(a_1) & \text{by } (*) \\
 \implies c_1 \leq \gamma(a_2) & \text{since } \gamma(a_1) = \gamma(a_2) \\
 \implies \alpha(c_1) \sqsubseteq \alpha(\gamma(a_2)) & \text{by monotonicity of } \alpha \\
 \implies a_1 \sqsubseteq \alpha(\gamma(a_2)) & \text{since } \alpha(c_1) = a_1 \\
 \implies a_1 \sqsubseteq \alpha(\gamma(a_2)) \sqsubseteq a_2 & \text{by (2)} \\
 \implies a_1 \sqsubseteq a_2 & \text{by transitivity of } \sqsubseteq
 \end{array}$$

Symmetrically, we can show that  $a_2 \sqsubseteq a_1$ . Thus, by antisymmetry of  $\sqsubseteq$ , we have  $a_1 = a_2$ , contradicting our assumption that they are distinct.

Therefore,  $\gamma$  is injective.

(c)  $\implies$  (a) :

Suppose, towards a contradiction, that there exists some  $a \in A$  such that  $\alpha(\gamma(a)) \neq a$ . We have

$$\begin{array}{ll}
 \alpha(\gamma(a)) \sqsubseteq a & \text{by (2)} \\
 \implies \gamma(\alpha(\gamma(a))) \leq \gamma(a) & \text{by monotonicity of } \gamma
 \end{array}$$

and,

$$\gamma(a) \leq \gamma(\alpha(\gamma(a))) \quad \text{by (1)}$$

and thus, by antisymmetry of  $\leq$ , we have  $\gamma(\alpha(\gamma(a))) = \gamma(a)$ . Since  $\gamma$  is injective, we have  $\alpha(\gamma(a)) = a$ , contradicting our assumption.

Therefore, for all  $a \in A$ , we have  $\alpha(\gamma(a)) = a$ .

Since we have shown that  $(a) \implies (b)$ ,  $(b) \implies (c)$ , and  $(c) \implies (a)$ , the three conditions are equivalent.

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3. State the condition for  $c = \gamma(\alpha(c))$  to hold for all  $c$ . When  $C$  is the set of sets of concrete states and  $A$  is a domain of static analysis, is it more reasonable to expect that  $c = \gamma(\alpha(c))$  or  $\alpha(\gamma(a)) = a$  to be satisfied, and why?

**Solution:** The condition for  $c = \gamma(\alpha(c))$  to hold is for  $\gamma$  to be surjective or  $\alpha$  to be injective as in the previous part.  $\gamma$  and  $\alpha$  would then be inverses of each other.

When switching to the domain of static analysis we lose information. Therefore, we would expect that  $\alpha(\gamma(a)) = a$ . However, having  $\gamma(\alpha(c)) = c$  would not make sense as it means that the abstract domain holds (as much, or) more information than the set of concrete states.

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**Exercise 2** (lub and glb). Let  $(A, \sqsubseteq)$  be a partial order such that every set  $S \subseteq A$  has a greatest lower bound. Either prove that every set  $S \subseteq A$  has a least upper bound, or show a counterexample.

**Solution:** Suppose we have some arbitrary set  $S \subseteq A$ , then we know that  $\sqcap S$  exists.

Let  $U$  be the set of all its upper bounds, i.e.  $U = \{x \mid \forall y \in S. y \sqsubseteq x\}$ . Since every subset of  $A$  has a greatest lower bound, we know that  $a = \sqcap U$  exists. Now we want to show that  $a$  is an upper bound on  $S$  and that it is the least one.

We want to show that  $\forall y \in S. y \sqsubseteq a$ .

Let  $L$  be the set of lower bounds on  $U$ , i.e.  $L = \{z \mid \forall x \in U. z \sqsubseteq x\}$ . Clearly,  $S \subseteq L$ , since  $U$  are the upper bounds on  $S$ . But,  $a$  is the greatest element in  $L$ , and thus  $\forall y \in S. y \sqsubseteq a$ . Thus,  $a$  is an upper bound on  $S$ .

Now take any upper bound  $x \in U$ . Since  $a = \sqcap U$ , we have  $a \sqsubseteq x$  and so  $a$  is the desired least upper bound.

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**Exercise 3** (Lattices). Consider the class of algebraic structures, called lattices, with signature  $(\vee, \wedge)$ , each of arity 2, satisfying the following axioms:

$$\begin{array}{ll|ll}
 x \vee y = y \vee x & (\vee\text{-Com}) & x \wedge y = y \wedge x & (\wedge\text{-Com}) \\
 (x \vee y) \vee z = x \vee (y \vee z) & (\vee\text{-Assoc}) & (x \wedge y) \wedge z = x \wedge (y \wedge z) & (\wedge\text{-Assoc}) \\
 x \vee x = x & (\vee\text{-Idem}) & x \wedge x = x & (\wedge\text{-Idem}) \\
 x \vee (x \wedge y) = x & (\vee\text{-Abs}) & x \wedge (x \vee y) = x & (\wedge\text{-Abs})
 \end{array}$$

1. Show that for any  $x$  and  $y$ ,  $x \wedge y = x$  if and only if  $x \vee y = y$ .

**Solution:** *Let  $x, y$  be two elements of the lattice.*

$(\implies) :$

$$\begin{aligned}
 & x \wedge y = x \\
 \implies & (x \wedge y) \vee y = x \vee y \\
 \implies & y \vee (x \wedge y) = x \vee y && \text{by } (\vee\text{-Com}) \\
 \implies & y \vee (y \wedge x) = x \vee y && \text{by } (\wedge\text{-Com}) \\
 \implies & y = x \vee y && \text{by } (\vee\text{-Abs})
 \end{aligned}$$

$(\impliedby) :$

$$\begin{aligned}
 & x \vee y = y \\
 \implies & x \wedge (x \vee y) = x \wedge y \\
 \implies & x = x \wedge y && \text{by } (\wedge\text{-Abs})
 \end{aligned}$$

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2. Define  $x \leq y$  by  $x \wedge y = x$ . Show that the lattice equipped with  $\leq$  is a partially ordered set.

**Solution:**

**Reflexivity:** *Follows immediately from  $(\wedge\text{-Idem})$*

**Antisymmetry:**

$$\begin{aligned}
 & x \leq y \leq x \\
 \implies & x \wedge y = x \text{ and } y \wedge x = y \\
 \implies & x \wedge y = x \text{ and } y \vee x = x && \text{by part 1} \\
 \implies & (y \vee x) \wedge y = x && \text{by substituting } x := (y \vee x) \\
 \implies & y \wedge (y \vee x) = x && \text{by } (\wedge\text{-Com}) \\
 \implies & y = x && \text{by } (\wedge\text{-Abs})
 \end{aligned}$$

**Transitivity:**

$$\begin{aligned} & x \leq y \leq z \\ \implies & x \wedge y = x \text{ and } y \wedge z = y \\ \implies & x \wedge y = x \text{ and } x \wedge (y \wedge z) = x \\ \implies & x \wedge y = x \text{ and } (x \wedge y) \wedge z = x && \text{by } (\wedge\text{-Assoc}) \\ \implies & x \wedge z = x \\ \implies & x \leq z \end{aligned}$$

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3. Show that  $\wedge$  and  $\vee$  are respectively the binary *greatest lower bound* and *least upper bound* for  $\leq$ .

**Solution:** Let  $x, y$  be two elements of the lattice. We will first prove that  $x \wedge y$  is a lower bound i.e.  $x \wedge y \leq x$  and  $x \wedge y \leq y$ . We will then show that it is the smallest among them, that is if  $z \leq x$  and  $z \leq y$  then  $z \leq x \wedge y$ .

$$\begin{aligned} & x \vee (x \wedge y) = x && \text{by } (\vee\text{-Abs}) \\ \implies & x \wedge (x \wedge y) = x \wedge y && \text{by part 1} \\ \implies & (x \wedge y) \wedge x = x \wedge y && \text{by } (\wedge\text{-Com}) \\ \implies & x \wedge y \leq x \end{aligned}$$

$$\begin{aligned} & y \vee (y \wedge x) = y && \text{by } (\vee\text{-Abs}) \\ \implies & y \vee (x \wedge y) = y && \text{by } (\wedge\text{-Com}) \\ \implies & y \wedge (x \wedge y) = x \wedge y && \text{by part 1} \\ \implies & (x \wedge y) \wedge y = x \wedge y && \text{by } (\wedge\text{-Com}) \\ \implies & x \wedge y \leq y \end{aligned}$$

Now, let  $z$  be an element of the lattice, such that  $z \leq x$  and  $z \leq y$ . We have

$$\begin{aligned} & z \leq x \text{ and } z \leq y \\ \implies & z \wedge x = z \text{ and } z \wedge y = z \\ \implies & (z \wedge x) \wedge y = z \\ \implies & z \wedge (x \wedge y) = z && \text{by } (\wedge\text{-Assoc}) \\ \implies & z \leq x \wedge y \end{aligned}$$

Therefore  $\wedge$  is the binary greatest lower bound for  $\leq$ . The proof for  $\vee$  is analogous. ◇

**Exercise 4.** Partitioning

1. Show that for any set  $S$ ,  $P_S = (2^S, \subseteq)$  is a lattice.

**Solution:** We can show that  $P_S$  is a complete lattice by showing that  $\subseteq$  is a partial order and that every subset of  $2^S$  has a greatest lower bound and a least upper bound. Reflexivity, antisymmetry, and transitivity of  $\subseteq$  can easily be checked by unfolding the definitions.

Given  $U \subseteq 2^S$ , we claim that  $\bigcup_{s \in U} s$  is the least upper bound of  $U$ .

Indeed we have  $\forall s' \in U, s' \subseteq \bigcup_{s \in U} s$  and for any  $b \in 2^S$ ,

$$(\forall s' \in U, s' \subseteq b) \leftrightarrow \bigcup_{s \in U} s \subseteq b$$

Analogously, we can prove that  $\bigcap_{s \in U} s$  is the greatest lower bound of  $U$ .

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2. Consider a set  $S = U \times V$ . For each set  $g \in 2^{U \times V}$ , define  $\bar{g} : U \rightarrow 2^V$  by  $\bar{g}(u) = \{v \in V \mid (u, v) \in g\}$ . Show that the bar function  $\bar{\cdot}$  defines a bijection between  $2^{U \times V}$  and  $U \rightarrow 2^V$ .

**Solution:** We show that the bar function is a bijection by defining another operator and proving that they are the inverse of each other:

$$\text{inv} : \bar{g} \mapsto \{(u, v) \in S \mid v \in \bar{g}(u)\}$$

For some  $g \in 2^{U \times V}$ :

$$\begin{aligned} \text{inv}(\bar{g}) &= \{(u, v) \in S \mid v \in \bar{g}(u)\} \\ &= \{(u, v) \in S \mid v \in \{v \mid (u, v) \in g\}\} \\ &= \{(u, v) \in S \mid (u, v) \in g\} \\ &= g \end{aligned}$$

Let  $f : U \rightarrow 2^V$ :

$$\begin{aligned} \overline{\text{inv}(f)} &= p \mapsto \{v \mid (p, v) \in \text{inv}(f)\} \\ &= p \mapsto \{v \mid (p, v) \in \{(p, v) \mid v \in f(p)\}\} \\ &= p \mapsto \{v \mid v \in f(p)\} \\ &= p \mapsto f(p) \\ &= f \end{aligned}$$

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3. Consider the set of all functions  $U \rightarrow 2^V$ . Define a lattice on this set that is isomorphic to  $(2^{U \times V}, \subseteq)$ .

**Solution:**

We define the following operators and relations on  $U \rightarrow 2^V$  and prove that  $\bar{\cdot}$  is a lattice isomorphism:

$$\begin{aligned} f_1 \sqcap f_2 &= p \mapsto f_1(p) \cap f_2(p) & f_1 \sqcup f_2 &= p \mapsto f_1(p) \cup f_2(p) \\ f_1 \sqsubseteq f_2 &\iff \forall p \in P. f_1(p) \subseteq f_2(p) \end{aligned}$$

In particular we have that  $f_1 \sqsubseteq f_2 \iff f_1 \sqcap f_2 = f_1$ .

We start by proving that  $\bar{\cdot}$  is a lattice homomorphism:

$$\begin{aligned} \overline{g_1 \cap g_2} &= p \mapsto \{v \mid (p, v) \in g_1 \cap g_2\} \\ &= p \mapsto \{v \mid (p, v) \in g_1 \wedge (p, v) \in g_2\} \\ &= p \mapsto \{v \mid (p, v) \in g_1\} \cap \{v \mid (p, v) \in g_2\} \\ &= p \mapsto \overline{g_1}(p) \cap \overline{g_2}(p) \\ &= \overline{g_1} \sqcap \overline{g_2} \end{aligned}$$

$$\begin{aligned} \overline{g_1 \cup g_2} &= p \mapsto \{v \mid (p, v) \in g_1 \cup g_2\} \\ &= p \mapsto \{v \mid (p, v) \in g_1 \vee (p, v) \in g_2\} \\ &= p \mapsto \{v \mid (p, v) \in g_1\} \cup \{v \mid (p, v) \in g_2\} \\ &= p \mapsto \overline{g_1}(p) \cup \overline{g_2}(p) \\ &= \overline{g_1} \sqcup \overline{g_2} \end{aligned}$$

We then derive  $\overline{g_1} \sqsubseteq \overline{g_2} \iff g_1 \subseteq g_2$  from the above equalities.

$$\begin{aligned} \overline{g_1} \sqsubseteq \overline{g_2} &\iff \overline{g_1} \sqcap \overline{g_2} = \overline{g_1} \\ &\iff \overline{g_1 \cap g_2} = \overline{g_1} \\ &\iff g_1 \cap g_2 = g_1 && \text{since } \bar{\cdot} \text{ is a bijection} \\ &\iff g_1 \subseteq g_2 \end{aligned}$$

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**Exercise 5** (post). Let  $S$  be any set,  $r \subseteq S \times S$  a binary relation, and  $I \subseteq S$ . Define  $post : 2^S \rightarrow 2^S$  by  $post(X) = I \cup r[X]$ . Prove that  $post$  is monotonic. Does  $post$  admit a least fixed point?

**Solution:** We first show that  $post$  is monotonic. Consider two arbitrary sets  $X, Y \subseteq S$  such that  $X \subseteq Y$ . We have:

$$\begin{aligned} X &\subseteq Y \\ \implies r[X] &\subseteq r[Y] \\ \implies I \cup r[X] &\subseteq I \cup r[Y] \\ \implies post(X) &\subseteq post(Y) \end{aligned}$$

Since  $(2^S, \subseteq)$  is a complete lattice, Tarski's fixed point theorem ensures the existence of a fixpoint for  $post$ .  $\diamond$

**Exercise 6.** Let  $f : [0, 1] \rightarrow [0, 1]$  such that

$$f(x) = \begin{cases} \frac{x^2}{4} + \frac{1}{10} & \text{if } x < \frac{1}{2} \\ \frac{1}{1+4^{-x}} & \text{if } x \geq \frac{1}{2} \end{cases}$$

Prove, using the fact that  $([0, 1], \leq)$  is a complete lattice, that there exists  $x \in [0, 1]$  such that  $f(x) = x$ .

**Solution:** The function  $f$  is monotonic (increasing) from  $[0, 1]$  to  $[0, 1]$ . By Tarski's fixed point theorem,  $f$  admits a least fixed point.  $\diamond$